AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 9 · STUDENT LESSON

Algorithms as Recipes

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Keep the current state visible.

Trace a score update, compare recipe order, diagnose a first error and run a fully specified paper robot.

Keep one habit: State the start → execute one instruction → record the new state.

MAIN LESSON · SECTION 1

Which value does the next instruction use?

> Trace current state, instruction order and the stopping point.

Start with two recipes

Two recipes use the same operations but produce different scores. Keep the starting state and every update so you can explain where their results begin to differ.

Retrieve the missing-feature limit

Two cards have the same recorded length but different widths and labels. Why can the length-only record fail to distinguish them?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can **trace** an ordered **algorithm** using the current value after every instruction.
- I can explain how changing the **initial state** or instruction order can change the result.
- I can locate the first incorrect update and distinguish copying a value from linking two **variable**s.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

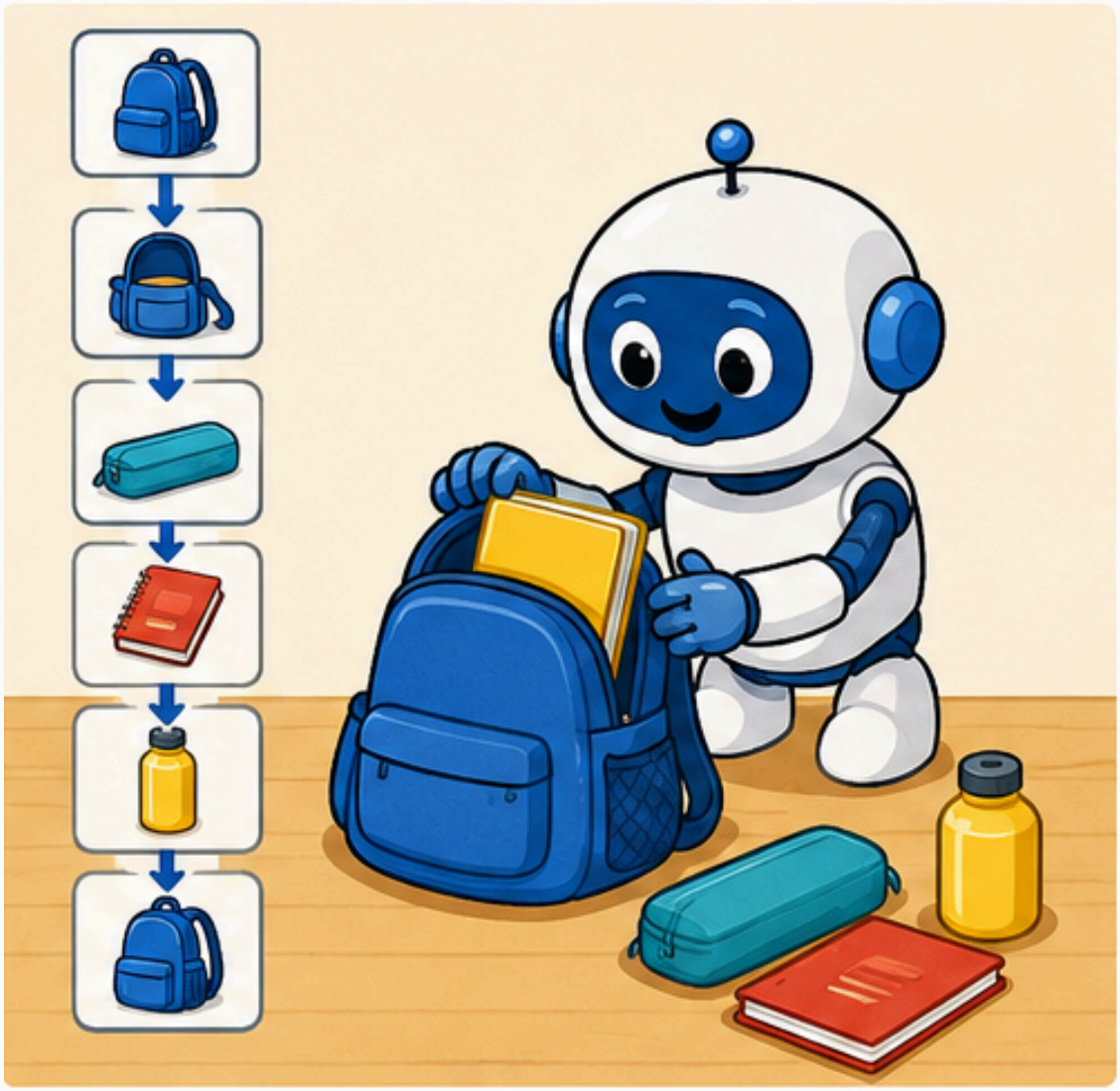
Remember before reading

Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

An **algorithm** is an ordered, testable procedure, with or without code.
</details>



LEARNING TARGET On numbered squares 0–5, start at 1 facing increasing numbers. Execute FORWARD 2, TURN, FORWARD 1, STOP. Define TURN as reversing direction. **Trace** position and direction after every instruction.

What if TURN were unspecified?

Learner name:

Readiness for the connected explanation

This week's end goal

By the end, I can **trace** changing **state** and explain why instruction order matters.

MAIN LESSON · SECTION 2

Track the current score, not just the start

> Use an erasable number box and a separate trace of earlier states.

Read the complete score procedure

A classroom score display starts with the number 4. Its instructions are: add 3 to score; double score; subtract 2 from score; report score and stop. What number should it report?

Before doing the arithmetic, name the value being tracked and write its starting value. Here the **variable** is score and the **initial state** is score 4.

Apply updates in their stated order

An **algorithm** gives precise instructions for a task. In this chapter each instruction is carried out once, in the written order. An update replaces the current value. After adding 3, score is 7.

The next instruction therefore doubles 7, not the original 4. Keeping only the starting value in mind is a common source of errors.

Separate the variable from the trace

A **variable** name stays the same while its value may change. Think of score as the label on an erasable number box. Updating the box does not keep all earlier numbers inside it.

A separate **trace** preserves those earlier **state**s so we can explain how the result was reached. This score procedure follows fixed arithmetic instructions; it does not learn from examples.

Predict the next update

- Name the variable and initial value. After the first instruction, which number should the doubling instruction read? Explain why the variable's name can stay unchanged while its value changes.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A paper robot cannot infer missing intentions; it needs ordered **step**s that another person can execute and test.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and **state** the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Record the state after every instruction

> An initial state plus a sequence of precise updates makes a run reproducible.

Use the latest value at each transition

Write the **initial state** first. For each instruction, read the latest value, perform the operation, and record the new **state**. A transition is the change from one recorded **state** to the next.

Follow the current score through three updates

STATE BEFORE	INSTRUCTION	STATE AFTER
Score is 4	Add 3 to score	Score becomes 7
Score is 7	Double score	Score becomes 14
Score is 14	Subtract 2 from score	Score becomes 12

Read the rows from top to bottom. Each row starts with the state produced by the previous row. Reporting the final score gives 12; reporting does not change the value.

Distinguish add from set

In words, double score means calculate twice its current value and store that result in score. Add 3 to score means increase the current value by 3. Set score to 3 means replace it with 3 instead.

These instructions sound similar but do different things: if score is 7, adding 3 produces 10, while setting it to 3 produces 3.

Compare runs from the same start

A **trace** is **evidence** of one execution from one **initial state**. To compare two procedures fairly, keep the **initial state** the same and list every intermediate value.

Then you can point to the first difference instead of only comparing the final numbers.

Explain a trace arrow

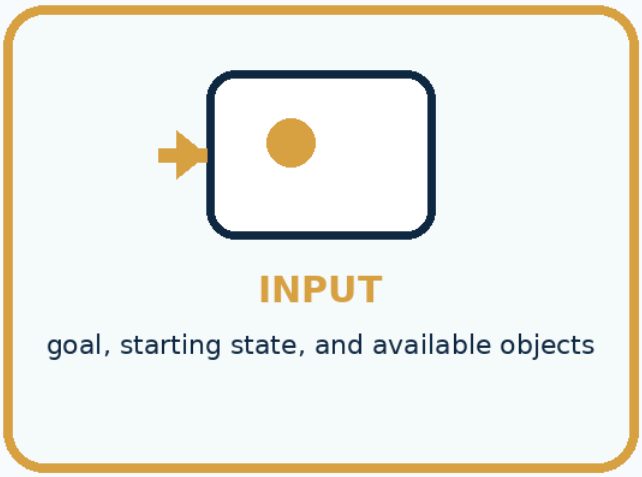
- Point to the **input**, the deciding **step** and the result in this week's visual. Explain the same **step** in ordinary words. A picture helps when you can say what each part represents.

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<summary>Earlier wording, alternatives and picture transcripts</summary>

VISUAL **MECHANISM**

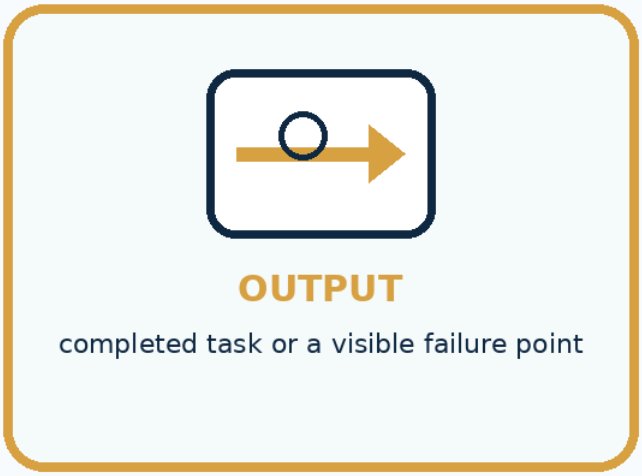
Input → Process → Output

Follow the information instead of imagining magic inside the machine.



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INPUT goal, starting **state**, and available objects

PROCESS follow ordered unambiguous **step**s

OUTPUT completed task or a visible failure point

CORE IDEA An **algorithm** is an ordered, testable procedure, with or without code.

UNDER THE HOOD An **algorithm** is a finite procedure whose **step**s and control flow can be executed. Correctness depends on **assumption**s, preconditions, order, and termination.

Code is one representation of an **algorithm**, not the definition itself.

Draw a fresh example with all three stages:

A closer explanation

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare instructions and ordering clues

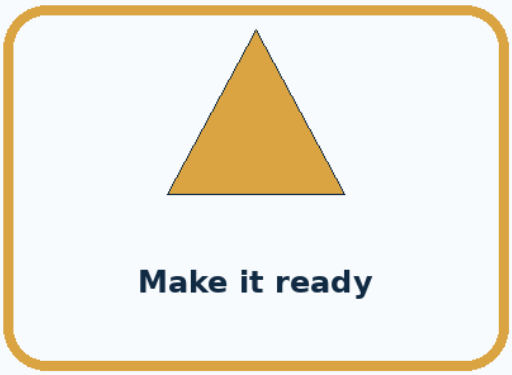
> Ask what each instruction makes the actor do and what must already be true.

Pictures paired with their evidence



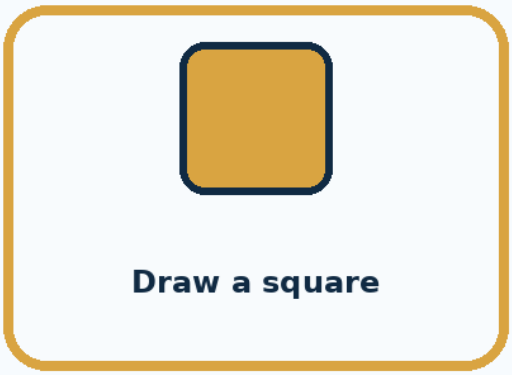
The case to investigate

CASE 1 — Pack a bag Clue: choose item, open bag, place item. Meaning: ordered task.



The case to investigate

CASE 2 — Make it ready Clue: meaning is un**state**d. Meaning: ambiguous instruction.



The case to investigate

CASE 3 — Draw a square Clue: four turns and moves. Meaning: testable procedure.



The case to investigate

CASE 4 — Choose the best book Clue: needs criteria. Meaning: goal without **algorithm**.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

BOUNDARY A goal is not yet an **algorithm**; each **step** must tell the actor what to do.

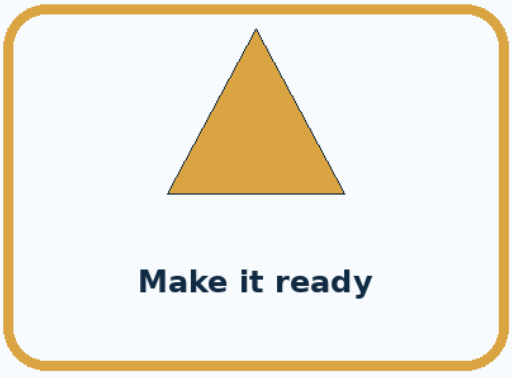
▼ Earlier wording, alternatives and picture transcripts

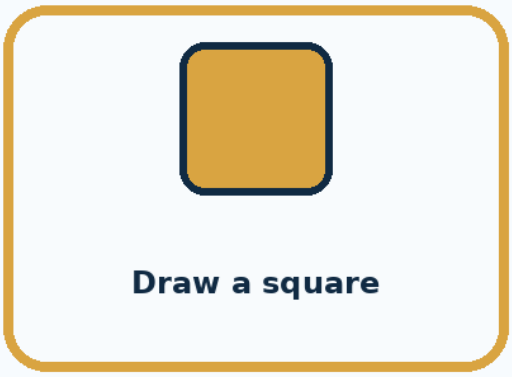
PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Pack a bag Clue: choose item, open bag, place item. Meaning: ordered task.

CASE 2 — Make it ready Clue: meaning is un**state**d. Meaning: ambiguous instruction.

CASE 3 — Draw a square Clue: four turns and moves. Meaning: testable procedure.

CASE 4 — Choose the best book Clue: needs criteria. Meaning: goal without **algorithm**.

Change one thing in the pictured case

Change: Change the starting score to 0 for both recipes.

Prediction: A reports 3 and B reports 7.

Reason: A: 0→0→4→3. B: 0→4→8→7.

Same operations. Different order.

> Start both recipes at 3 and keep the intermediate states visible.

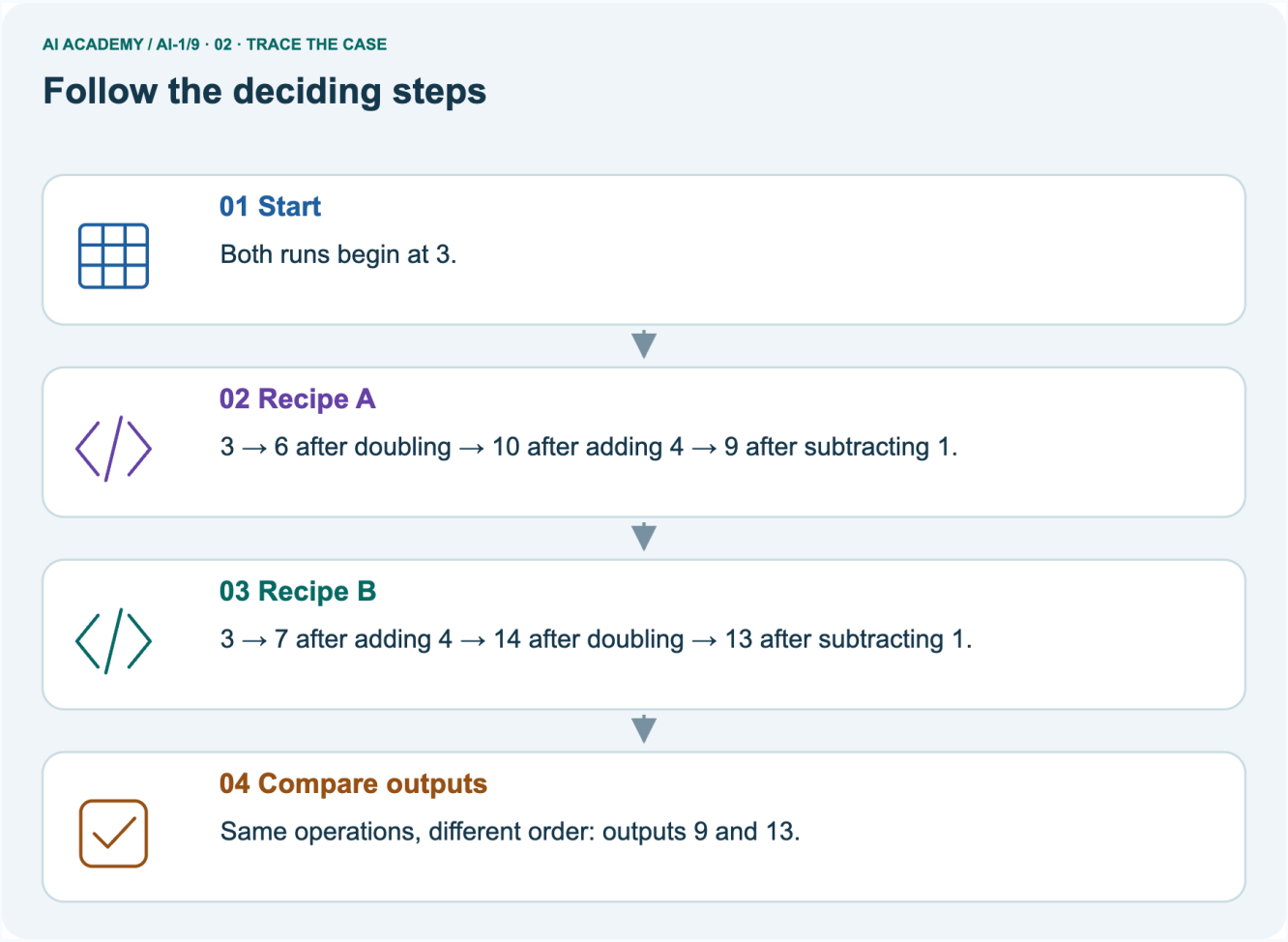
The two complete recipes

A fictional paper badge machine transforms an initial score using an ordered recipe.

Start score=3. Recipe A: double, add 4, subtract 1, report.

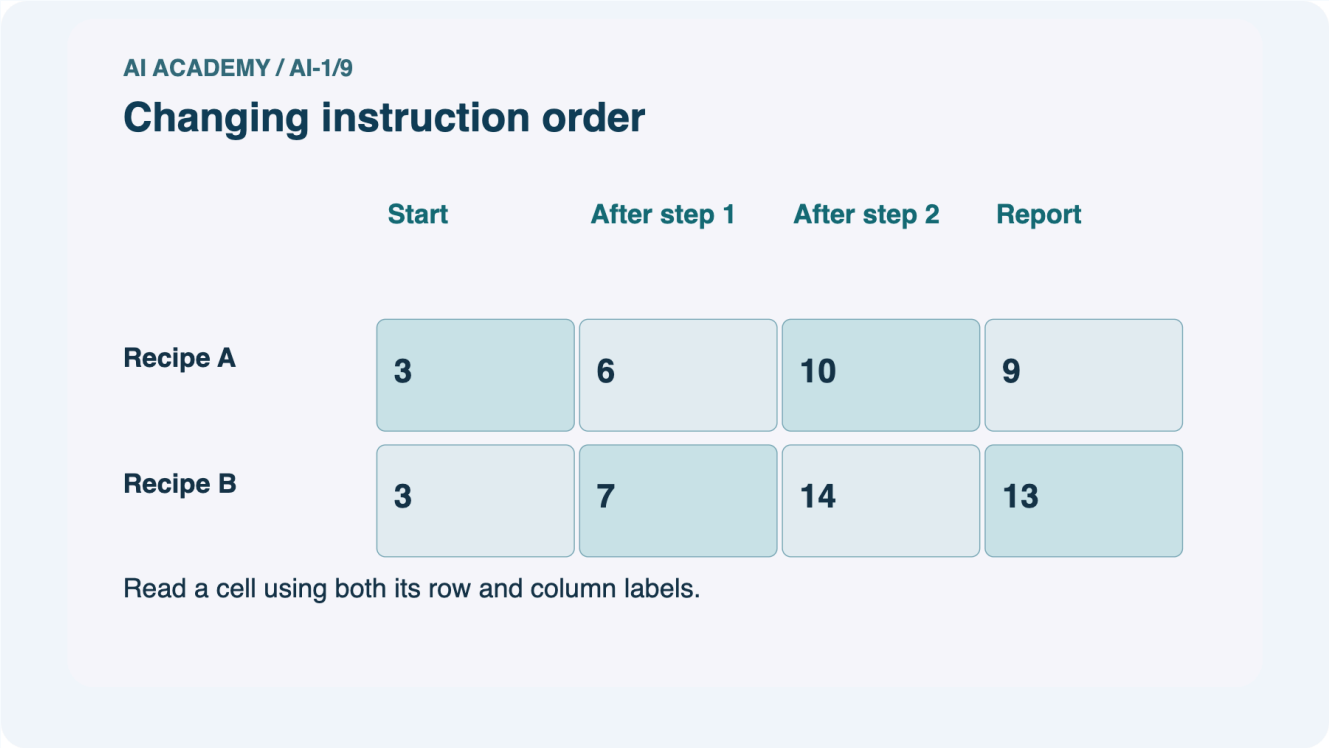
Recipe B swaps the first two operations but keeps the same starting score.

Follow both ordered traces



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare intermediate values and outputs



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the final difference

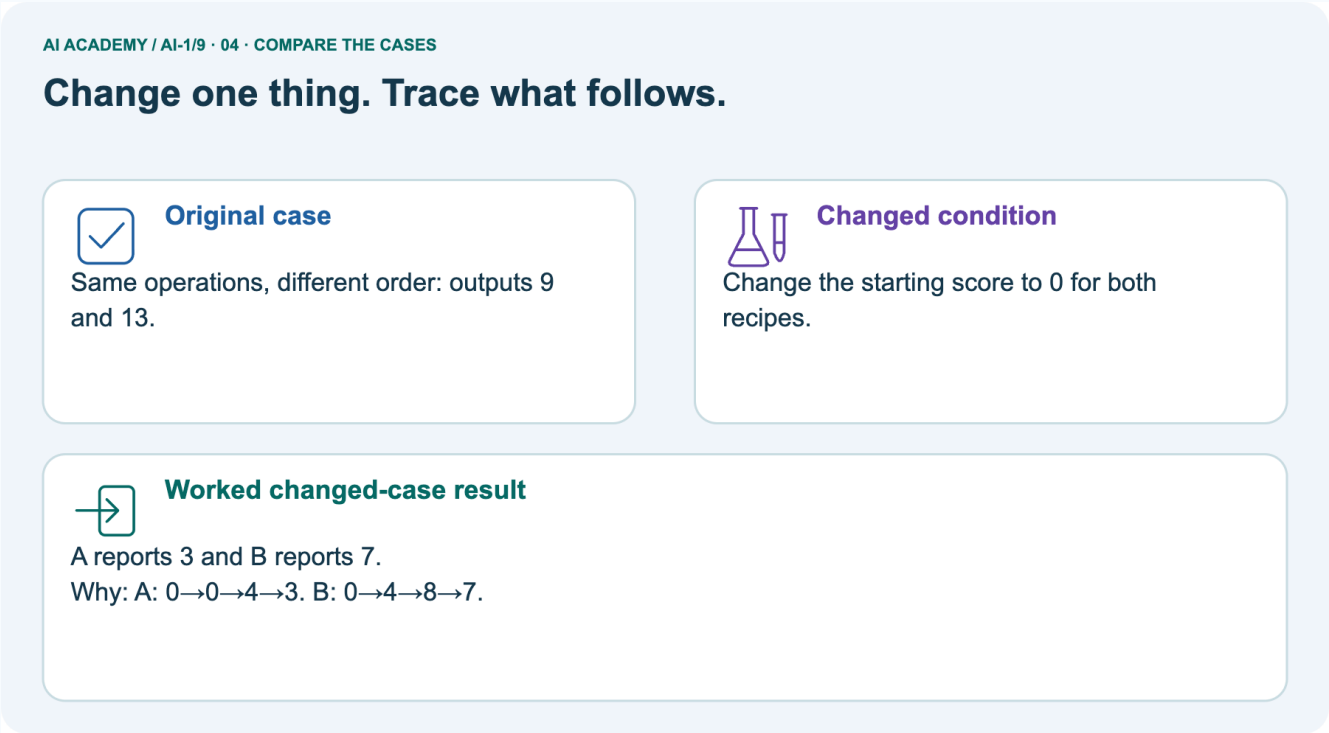
Every instruction reads the current score, not the original one unless explicitly told to. In B, doubling also doubles the added 4. In A, the extra 4 is added after doubling.

The final difference of 4 follows from that order, not from different starting data.

Predict from a new starting value

- Before checking, start both recipes at 0. Keep each recipe's instruction order fixed. Write the start and every update; mark the first transition where the two runs differ.

▼ Check the zero-start traces



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit what one run verifies

One **trace**d **input** checks one run; it does not establish every recipe is equivalent or validate an entire program.

Locate the first changed state

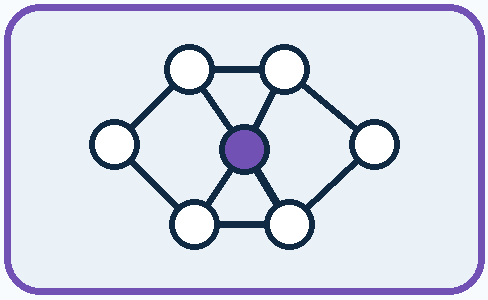
- Read the labelled picture: Use the arrows to explain where the two zero-start runs first differ.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every **step** connects a claim to observable **evidence**.



CASE A paper robot cannot infer missing intentions; it needs ordered **step**s that another person can execute and test.

STEP 1 — OBSERVE goal, starting **state**, and available objects

STEP 2 — **TRACE** follow ordered unambiguous **step**s

STEP 3 — CONCLUDE completed task or a visible failure point

MODEL REASONING A strong **algorithm state**s the starting arrangement, names each object, orders actions, and includes checks such as whether the bag is open.

The revision targets the failed **assumption** rather than replacing every **step**.

DO NOT OVERCLAIM A goal is not yet an **algorithm**; each **step** must tell the actor what to do.

What new **evidence** could change this answer?

A closer explanation

Stage from score 4	Add then double	Double then add
After first instruction	Add 3 gives 7	Double gives 8
After second instruction	Double gives 14	Add 3 gives 11
After subtracting 2	Report 12	Report 9

Both procedures use add 3, double and subtract 2, but the first two operations are exchanged. Doubling after the addition also doubles the extra 3. Adding afterward does not.

The final difference comes from the order, even though both procedures started at 4.

Now use two number boxes: A starts at 2 and B starts at 0. Copy the current value of A into B, then add 5 to A. After the copy, A is 2 and B is 2. After the addition, A is 7 and B remains 2.

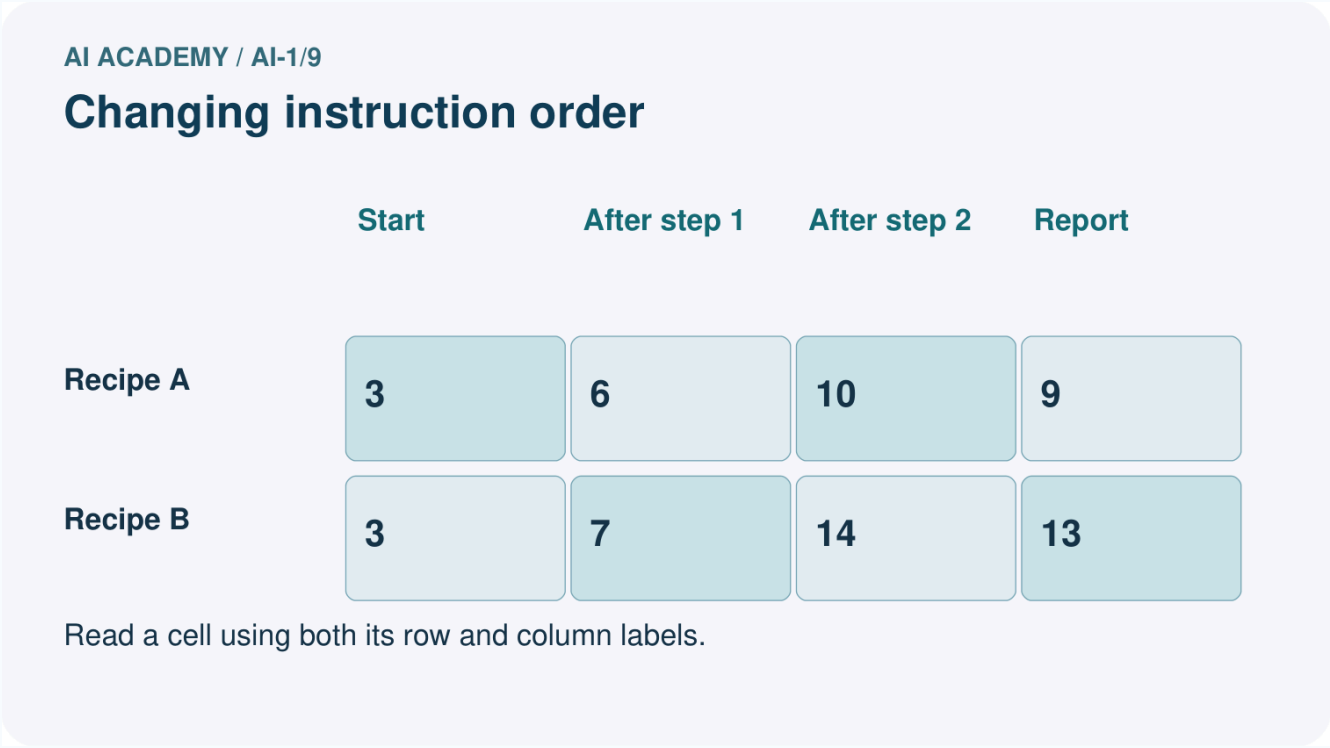
Copying stores the value at that moment; it does not create a lasting connection between these number boxes. Record both values after each instruction, including the one that did not change.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one **step** at a time. Then uncover the result and explain your first mismatch. Ask why a **step** is allowed, not only what number it produces.

A pictorial worked case: Algorithms as Recipes

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/9

Algorithms as Recipes

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Start

Both runs begin at 3.



2. Recipe A

3 -> 6 after doubling -> 10 after adding 4 -> 9 after subtracting 1.



3. Recipe B

3 -> 7 after adding 4 -> 14 after doubling -> 13 after subtracting 1.



4. Compare outputs

Same operations, different order: outputs 9 and 13.

CHANGE ONE THING

Change the starting score to 0 for both recipes.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Start	Both runs begin at 3.
2. Recipe A	3 → 6 after doubling → 10 after adding 4 → 9 after subtracting 1.
3. Recipe B	3 → 7 after adding 4 → 14 after doubling → 13 after subtracting 1.
4. Compare outputs	Same operations, different order: outputs 9 and 13.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Change the starting score to 0 for both recipes. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Matching outputs can hide different procedures

> A final answer alone does not show which steps were followed.

Compare equal outputs with different intermediate states

A final value does not reveal every **step** that produced it. Starting at 10, adding 2 then subtracting 2 reports 10. Adding 4 then subtracting 4 also reports 10. The intermediate **state**s differ.

If a task requires a particular procedure, the final answer alone cannot establish that the required **step**s were followed.

Specify the initial state and stopping point

Before running an **algorithm**, **state** its initial values, the meaning and order of each instruction, and when to report and stop. Our short procedures stop after their listed **step**s.

A correct **trace** shows what those instructions do for the given start; it does not establish that the procedure solves every possible task or that a real-world display was measured correctly.

Repair and explain

- CLAIM A People will know what I meant.
- CLAIM B An **algorithm** must be written in a programming language.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

Two learners use the same instruction list but report different final scores. Name two things to compare before deciding which trace is wrong, and explain how a trace helps

▼ Earlier wording, alternatives and picture transcripts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Pack a bag	choose item, open bag, place item	ordered task
Make it ready	meaning is unstated	ambiguous instruction
Draw a square	four turns and moves	testable procedure
Choose the best book	needs criteria	goal without algorithm

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: repair the first incorrect update

> Continue from the corrected current state rather than changing only the last answer.

Diagnose the trace 4,7,8,6

A learner **trace**s the score procedure as 4, 7, 8, 6. The first update is correct: 4 plus 3 is 7. The first error is the next update: double should use the current 7 and produce 14, but the learner doubled the starting 4.

Correct that **step** and recompute the later subtraction, giving 12. Changing only the final answer would hide the faulty reasoning.

Write your new score trace

> **Start score at 5. Add 2 to score, double score, then subtract 1. Write the initial state and every later state. Explain which value is doubled and why**

Compare starting states before blaming an instruction

If two **trace**s disagree, compare the **initial state** before blaming an instruction. If the starting values match, compare the **state**s in order and stop at the first difference.

Check the instruction used there, repair it if needed, and continue from the corrected **state**.

Use W2 to distinguish replacement and addition

Use workbook W2 for the connected example "**Algorithm**s and changing **state**". First fill the **input** and rule boxes. Try the middle **step**s before asking for a hint. After a hint, try the next **step** yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your **trace** to a partner. Your partner points to one **step** to check. Swap roles on the next case. Each person writes their own explanation.

Challenge “every instruction uses the original value”

- Use the worked case for **Algorithm**s and changing **state**. Proposed explanation: “Every instruction uses the original number”. Decide whether this explanation is supported.

  Point to the **step** or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which **step** you need help with.

Explain the zero-start picture

Use the arrows to explain where the two zero-start runs first differ.

My explanation using a pictured step and the given facts

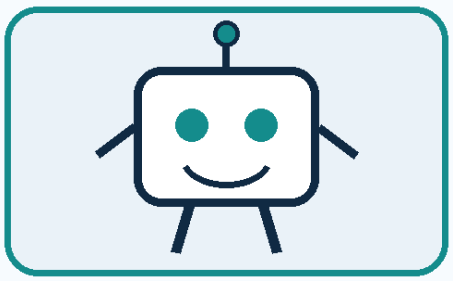
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: An **algorithm** is an ordered, testable procedure, with or without code.

Use a fresh example to explain **algorithm**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: An **algorithm** must be written in a programming language.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Distinguish replacement from addition

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate: give a paper robot executable directions

> Use stated command meanings and initial conditions so a partner can reproduce or challenge the result.

Your initial state and command rules

Use a paper bag-state card and pencil/book tokens. Start with bag=CLOSED, contents=empty, and both tokens on the mat. OPEN changes CLOSED to OPEN. PUT(item) requires an open bag and that token on the mat, then moves it into the bag. CLOSE changes OPEN to CLOSED. REPORT lists the contents and stops. If a requirement fails, STOP and name the failed requirement; do not invent a successful move.

Predict, execute on paper and compare

1. Predict and trace OPEN, PUT(pencil), PUT(book), CLOSE, REPORT. Record the bag state, contents and remaining mat tokens after each instruction. Then reset to the same start and try PUT(book), OPEN, PUT(pencil), CLOSE, REPORT. Have a partner act exactly as written. Identify the first failure and rewrite the order before a fresh trace.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep the trace and first failure

Case	Prediction	Observed result	What it means
Normal			
Edge			

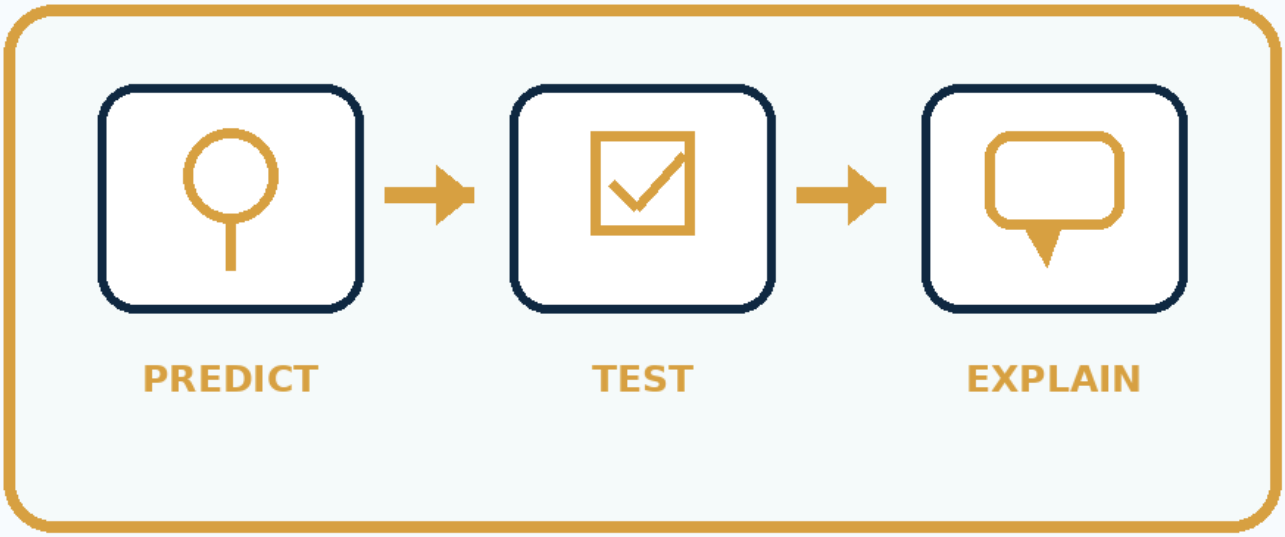
Before trying: what do I expect, and why? While trying: which **step** could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only **trace**d the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Paper Robot Bag Pack

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Write directions for a paper robot to pack a bag.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: trace two changing number boxes

> Track both X and Y after every instruction, including values that stay unchanged.

The new procedure and requested swap

For a new display, X starts at 4 and Y starts at 1. Copy the current X into Y; add 3 to X; double Y; report both and stop. Record X and Y initially and after each instruction. Explain why Y does not become 7 when X changes.

Then swap the first two instructions and **trace** the new procedure from the same **initial state**.

My independent explanation, trace and limits

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported **step** and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Trace two changing number boxes

Week 9 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Instructions say pick up a pencil, write your name, then find paper. Find the first ordering problem and rewrite the three **step**s so another learner can follow them.

My answer, with a **trace**, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Write the first executable sorting **algorithm** for one easy item before adding branches or AI.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

On numbered squares 0–5, start at 1 facing increasing numbers. Execute FORWARD 2, TURN, FORWARD 1, STOP. Define TURN as reversing direction. **Trace** position and direction after every instruction. What if TURN were unspecified?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Write your sorter’s first reproducible procedure

> Show each state update before adding more branches or a learned model.

This week’s project decision

THIS WEEK'S CHECKPOINT Write the first executable sorting **algorithm** for one easy item before adding branches or AI.

Write the sorter’s **step**s in order, showing which values each **step** updates so a partner can reproduce the result.

Plan information, tests and review

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Keep a goal separate from executable steps

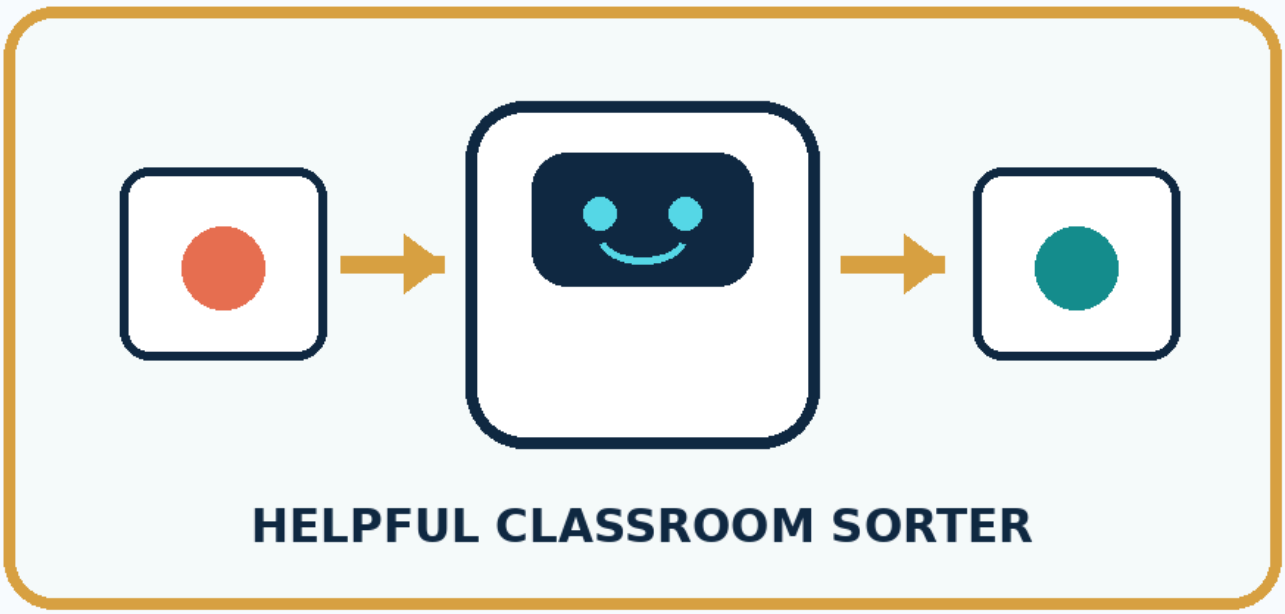
RESPONSIBLE DESIGN A goal is not yet an **algorithm**; each **step** must tell the actor what to do.

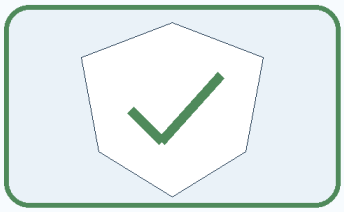
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: explain why a transition follows

> Close your notes and name the current value, operation and resulting state.

Your exit explanation

- Close your notes. Explain this goal in your own words: **Trace** an ordered **algorithm** using the current value after every instruction

> **My deciding step, evidence and one remaining question**
>
> | |
> |---|
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Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or **assumption** and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
algorithm	ordered procedure for a task	
step	one executable action	
sequence	the order in which steps run	
assumption	condition taken for granted	

Terms for the connected explanation

Word	Meaning
Variable	A named place for a value that can be updated
State	The current values being tracked at a particular point in a process
Initial state	The values before the first instruction is carried out

Word	Meaning
Trace	A record of the state as instructions are carried out in order

Repair a claim using a word

- CLAIM TO REPAIR People will know what I meant.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

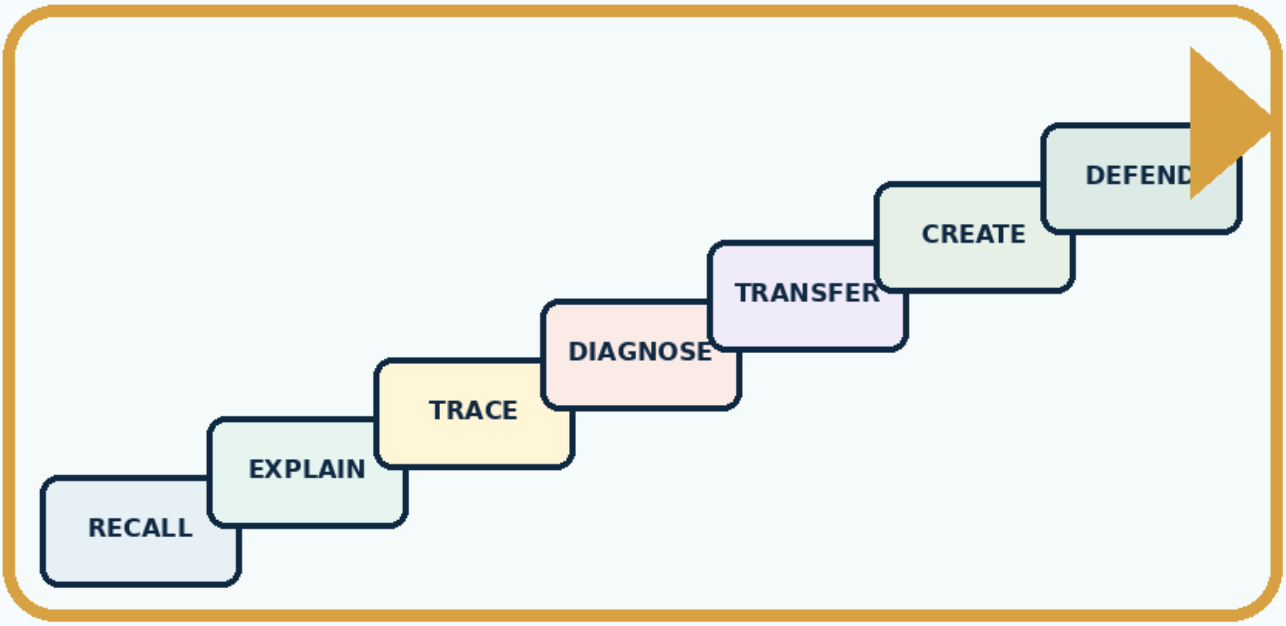
- RECALL Define **algorithm**.
- EXPLAIN An **algorithm** is an ordered, testable procedure, with or without code.
- **TRACE** Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: People will know what I meant.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each **step** asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Trace two changing number boxes is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ **Trace input**, **process**, and **output**.

- ☐ **State** the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Trace two changing number boxes is the selected independent case.



EXPLANATION CHECK State the initial values, update the current state in order, keep unchanged values visible and find the first differing transition.

Independent **evidence**:

SELF-CHECK A strong **algorithm state**s the starting arrangement, names each object, orders actions, and includes checks such as whether the bag is open.

The revision targets the failed **assumption** rather than replacing every **step**.

NEXT BRIDGE Carry today's **evidence** habit into: If/Then Decisions.

▼ **Extra practice: Trace a robot's position**

REFERENCE · SECTION 13

Extra practice: Trace a robot's position

> Try the question before opening its worked explanation.

Read the supplied case and question

A robot starts on the left of three squares, facing right. It executes MOVE, MOVE, STOP, each MOVE being one square.

Trace the position after each command.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Middle, rightmost, unchanged. Direction, distance and starting conditions must be specified.

Evidence or stage	Value or result
Start	Left square
MOVE	Middle square
MOVE	Right square
STOP	Right square

▼ Earlier wording, alternatives and picture transcripts

Week 9 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Trace a robot's position

A robot starts on the left of three squares, facing right. It executes MOVE, MOVE, STOP, each MOVE being one square.

Trace the position after each command.

Middle, rightmost, unchanged. Direction, distance and starting conditions must be specified.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Trace an ordered algorithm using the current value after every instruction	Explain the deciding step, show a trace or test, and state one limit.
Explain how changing the initial state or instruction order can change the result	Explain the deciding step, show a trace or test, and state one limit.
Locate the first incorrect update and distinguish copying a value from linking two variables	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional paper badge machine transforms an initial score using an ordered recipe.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional paper badge machine transforms an initial score using an ordered recipe.
- Which labelled **step** matches this detail? Same operations, different order: **output**s 9 and 13.
- 1. Recipe A
- 2. Compare **output**s
- 3. Recipe B
- 4. Start
- Hint if needed: Read the **step**s in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported explanation: Compare **output**s Same operations, different order: **output**s 9 and 13. Every instruction reads the current score, not the original one unless explicitly told to.

In B, doubling also doubles the added 4. In A, the extra 4 is added after doubling. The final difference of 4 follows from that order, not from different starting data.

Changed-case question

- Changed-case practice: Change the starting score to 0 for both recipes. Which conclusion is supported by the supplied facts?
- 1. A reports 3 and B reports 7.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding **step** in the pictorial **trace**. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: A reports 3 and B reports 7. A: 0→0→4→3. B: 0→4→8→7.

Repair and save your work

Review Features: Useful Clues (ai-1/8). Goal: Choose recorded properties that preserve distinctions needed for a **state**d task.

Short practice: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.. Return to this week and explain what you changed.

Review Sound Becomes a Signal (ai-1/6). Goal: Interpret an ordered **sequence** of sound-signal samples with a **state**d time interval. Short practice: Does the all-zero retained record prove the full signal was silent?

Use the picture.. Return to this week and explain what you changed.

Review Sensors Are Messengers (ai-1/3). Goal: Describe a measurement using its quantity, value, unit and conditions. Short practice: Why can we calculate the earlier gap but not predict an exact new B reading?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

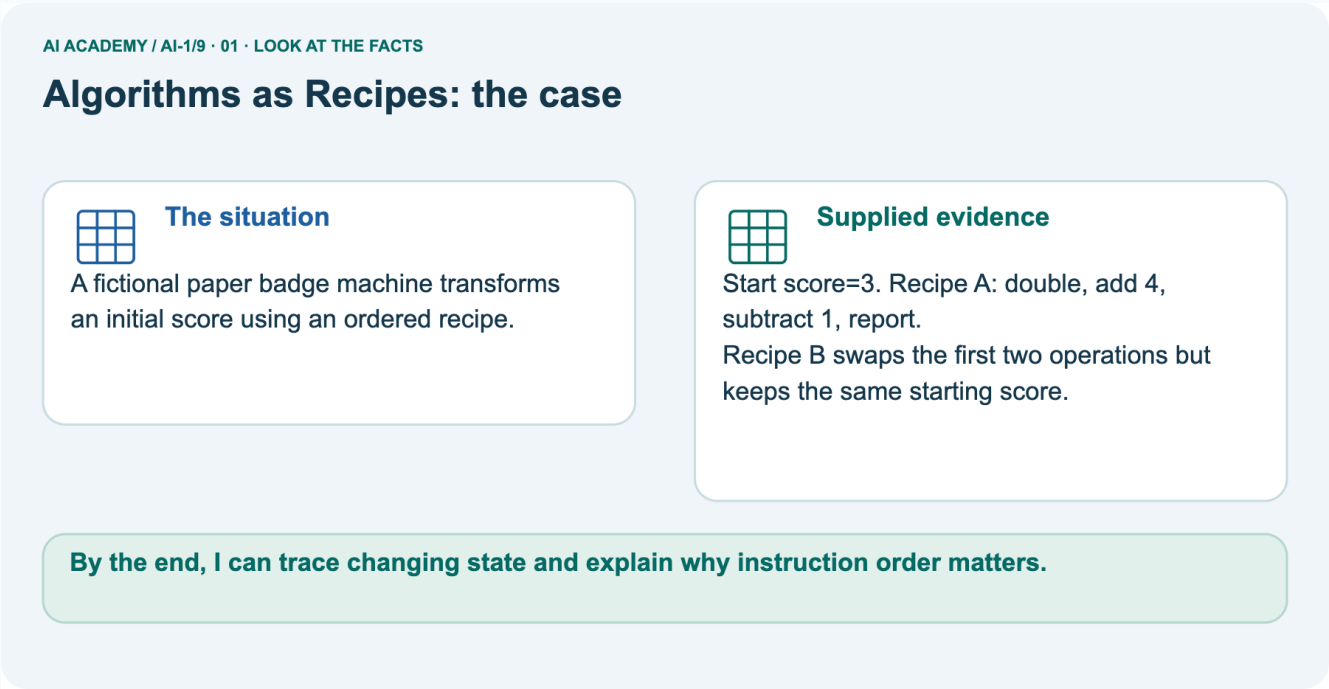
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Algorithms as Recipes: the case



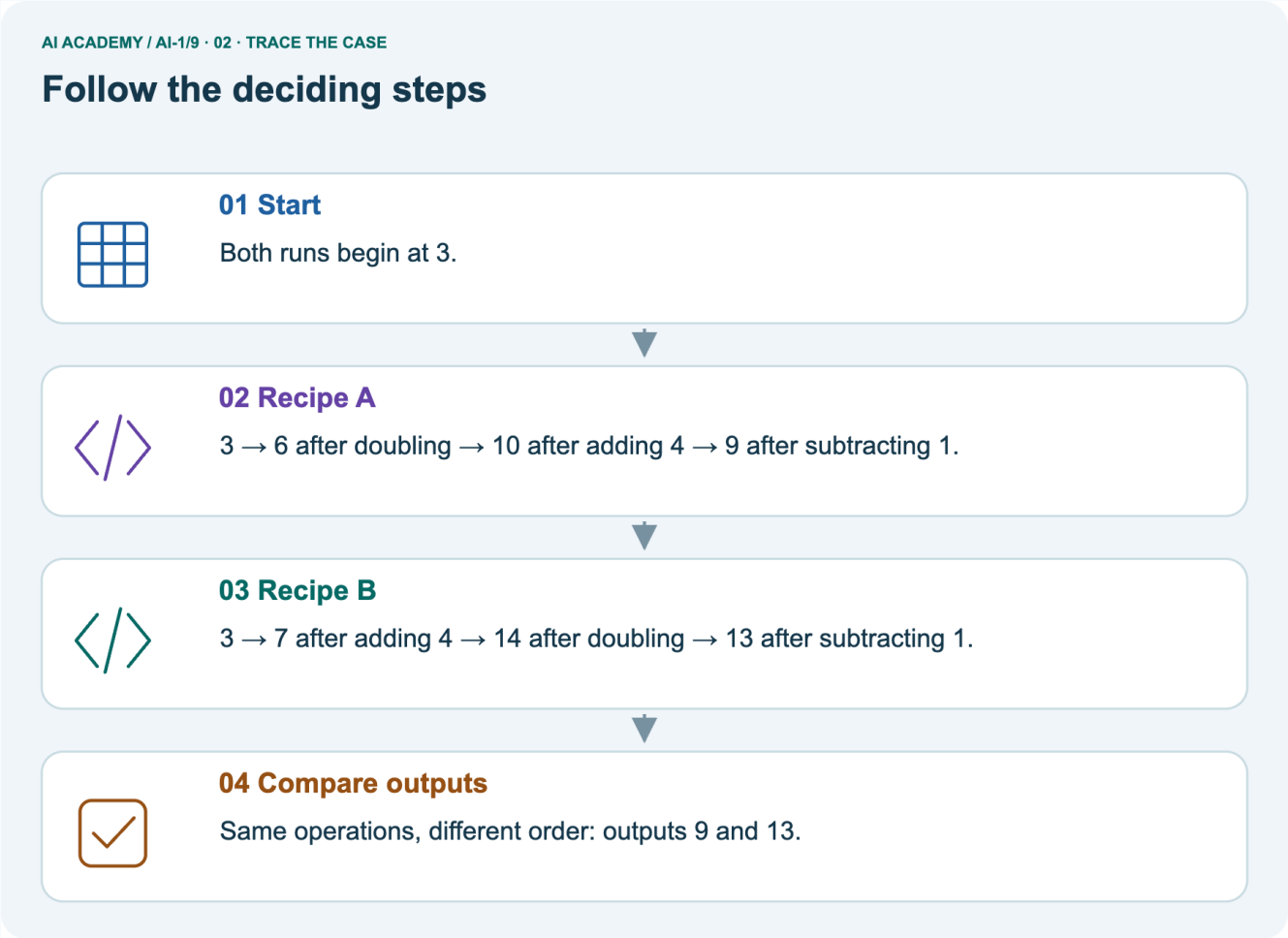
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional paper badge machine transforms an initial score using an ordered recipe. Supplied facts: Start score=3. Recipe A: double, add 4, subtract 1, report.

Recipe B swaps the first two operations but keeps the same starting score.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

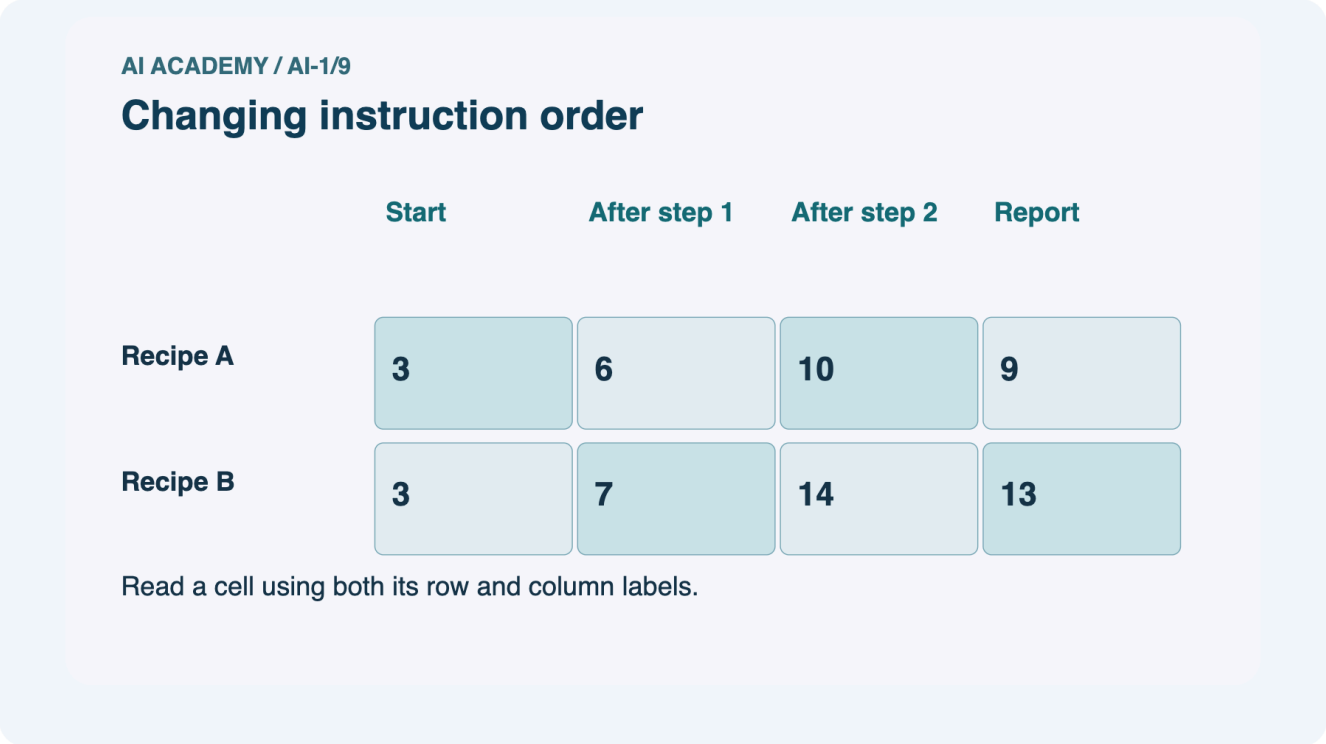


Follow the labelled picture with the matching explanation. Open the vector diagram

Start: Both runs begin at 3. → Recipe A: 3 → 6 after doubling → 10 after adding 4 → 9 after subtracting 1. → Recipe B: 3 → 7 after adding 4 → 14 after doubling → 13 after subtracting 1. → Compare **output**s: Same operations, different order: **output**s 9 and 13.

Point to each numbered stage and explain which recorded **input** it uses.

03. Changing instruction order

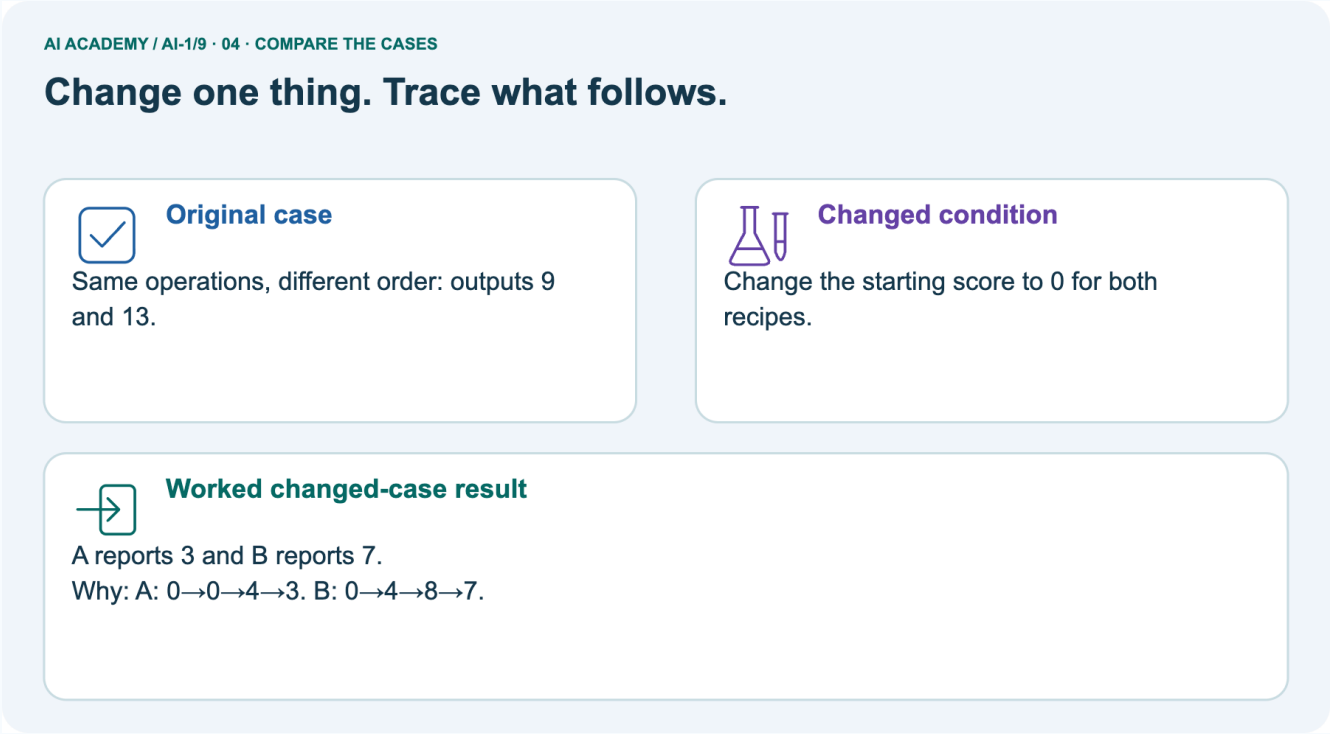


Follow the labelled picture with the matching explanation. Open the vector diagram

Changing instruction order. Recipe A: Start = 3, After **step** 1 = 6, After **step** 2 = 10, Report = 9. Recipe B: Start = 3, After **step** 1 = 7, After **step** 2 = 14, Report = 13.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.

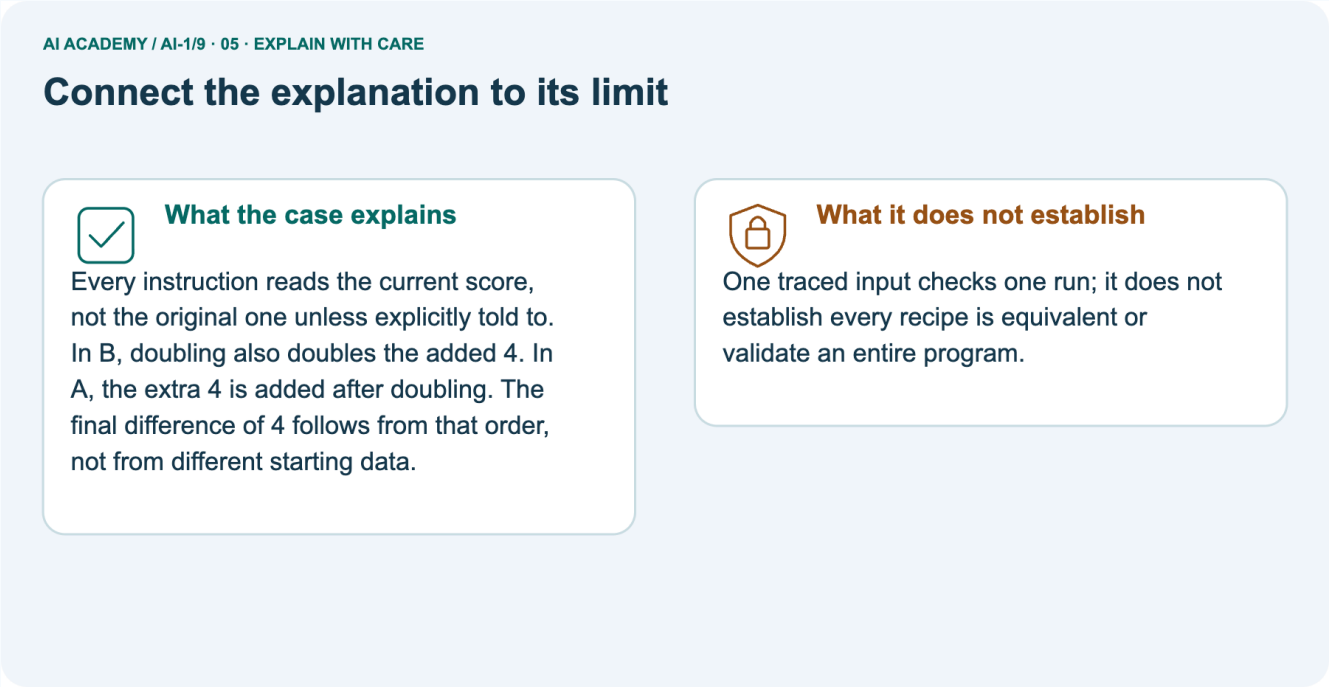


Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Same operations, different order: **output**s 9 and 13. Changed condition: Change the starting score to 0 for both recipes. Expected result: A reports 3 and B reports 7. Reason: A: 0→0→4→3. B: 0→4→8→7.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



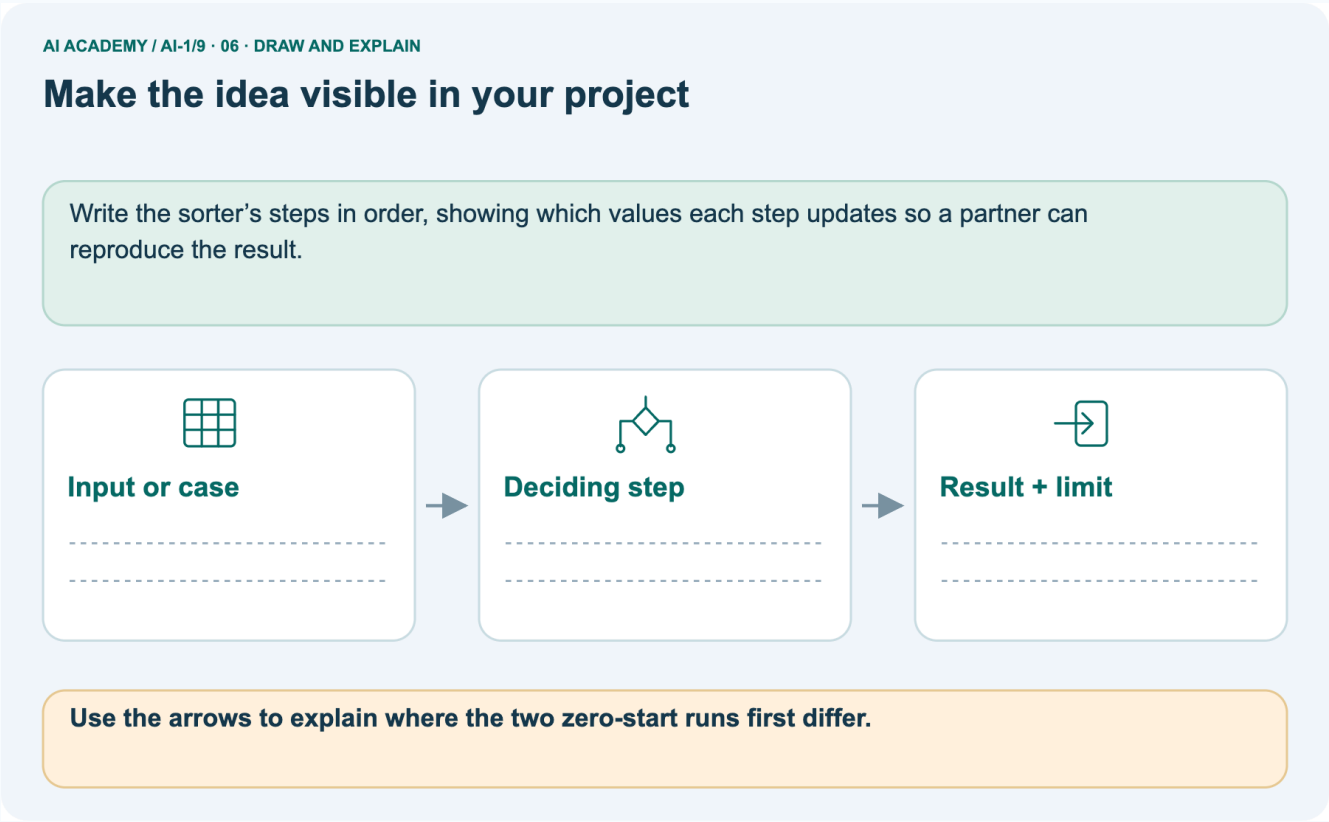
Follow the labelled picture with the matching explanation. Open the vector diagram

Every instruction reads the current score, not the original one unless explicitly told to. In B, doubling also doubles the added 4. In A, the extra 4 is added after doubling.

The final difference of 4 follows from that order, not from different starting data. Limit: One **trace**d **input** checks one run; it does not establish every recipe is equivalent or validate an entire program.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Write the sorter's **step**s in order, showing which values each **step** updates so a partner can reproduce the result. Draw your **input**, deciding **step**, result and limit.

Supported question: Use the arrows to explain where the two zero-start runs first differ.

Draw your project **trace** and justify the deciding **step** using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Algorithms as Recipes: the case

02. Follow the deciding steps

03. Changing instruction order

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.